

Michael Psenka

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EDUCATION

Princeton University, Class of 2021

Bachelor's in **Mathematics**, certificates in **Applications of Computing** and **Applied Math**

- Undergrad CS coursework: *Algorithms and Data Structures, Programming Systems, Reasoning about Computation, Machine Learning and Pattern Recognition, Mathematical Machine Learning*
- Graduate CS coursework: *Deep Learning Theory, Complexity Theory, Information Theory, Reinforcement Learning*
- Math coursework: *Partial Differential Equations (Geometric PDE), Stochastic Differential Equations, Real Analysis (Integration Theory), Complex Analysis, Linear Algebra, Group Theory, Ring Theory, Representation Theory, Commutative Algebra, Probability & Stochastic Systems*
- GPA: 3.6

AI RESEARCH

Lottery ticket & pruned neural networks

Princeton, Nov. 2020

- Worked in a team of 4 to methodically study the expressivity and data dependency of various pruning methods for deep neural networks
- Established new findings and conclusions on important limitations for pruning methods at initialization, leading to deeper insight on the mechanisms behind pruning methods as well as inspiration for future improvement

Multi-view point cloud reconstruction

Stanford (Guibas Lab), June 2020 – Present

- Working with Tolga Birdal on a novel approach to multi-view reconstruction in computer vision that completely bypasses the pairwise view registration step, which can be noisy and expensive
- Involves novel methods in non-Euclidean machine learning, optimal transport

APPLIED MATHEMATICS RESEARCH

Reduced order modeling

Princeton, Sept. 2020 – Present

- Working with Prof. Weinan E to study reduced order modeling from a theoretical and continuous perspective, encapsulating modern methods such as LSTM and transformers
- Employs techniques from dynamical systems theory and functional analysis

Non-convex optimization over tensor manifolds

Princeton, June 2019-2020

- Worked with Prof. Nicolas Boumal to study second-order methods on problems over low-dimensional manifolds in high-order tensor spaces
- Found, proved, and coded a new second-order method over the Tensor Train format that provably beats the state-of-the-art in certain poorly conditioned problems

PURE MATHEMATICS RESEARCH

Spectral Geometry

Princeton, Mar. – Sept. 2020

- Worked in a group of 4 undergraduates under Prof. Javier Gómez-Serrano to study an open problem: proving the conjecture that the ratio of the first two eigenvalues of a triangle is maximized by the equilateral triangle
- Extended and improved upon previous analytic bounds on the problem, leading to a complete proof and solving the open problem

PUBLICATIONS & WORKSHOP PROCEEDINGS

Second-order optimization for tensors with fixed tensor-train rank. **Michael Psenka**, Nicolas Boumal (1st author). [Link](#)

- Accepted to the OPT2020 NeurIPS workshop

Reconstruction without Registration. **Michael Psenka**, Tolga Birdal (1st author). [Link](#)

- Accepted to the IROS2020 geometric methods workshop

A Proof of The Triangular Ashbaugh-Benguria-Payne-Pólya-Weinberger Inequality. Ryan Arbon, Mohammad Mannan, **Michael Psenka**, Seyoon Ragavan (co-1st author). [Link](#)

- To appear in Journal of Spectral Theory

AWARDS

Peter A. Greenberg '77 Memorial Prize

June 2020

- Awarded for outstanding accomplishments in Mathematics by juniors

Manfred Pyka Memorial Prize*June 2018*

- Given to outstanding Physics undergraduates who have shown excellence in course work and promise in independent research

HackPrinceton First Place*April 2018*

- *Won first place* for developing A.I.D.A.N. at HackPrinceton Spring 2018
- A.I.D.A.N. is a chatbot that, once you upload your dataset, has the capacity to do useful statistical analysis and machine learning on your dataset through voice commands. [Link](#)

EXTRACURRICULARS

- **Princeton Pianist Ensemble**
 - *Publicist*: 2018
 - Group of around 40 pianists that perform school-wide bi-annually, with an emphasis on collaborative performance
 - Participated in performances for charity, e.g. showcasing for funded arts program for low-income schools
 - Performed in a virtual concert during quarantine. [Link to video](#)
- **Math Club**
- **Princeton Data Science**

EMPLOYMENT**Princeton University Department of Mathematics***June – Sept. 2019*

- Research under Prof. Nicolas Boumal (see non-convex optimization over tensor manifolds above) was funded by the National Science Foundation through award DMS-1719558

Moovila Inc.*June – Aug. 2018, '17, '16*

- Worked closely with the dev team, participating in stand-up and sprints regularly
- Developed a machine learning algorithm for workplace analytics
- Mathematically modeled collision avoidance in network analysis animation
- Worked through a patent application for proprietary software
- Worked on improving the search engine for quicker and more robust search results
- Denormalized relational database to NoSQL, maximizing data access efficiency and cost-efficiency

TECHNOLOGIES

- **Currently using w/ proficiency**: Python (PyTorch, NumPy), MATLAB
- **Extensive previous experience**: JavaScript, Java, C#, C
- **Previous experience**: AWS, Unix/Linux, Web3, HTML, CSS